Improving map quality: Airy + FAST-LIO2 handheld twin (research)

2026-10-21. Sources are linked inline. ***[U]*** *marks items that are unverified or inferred.*

Airy driver, timing and IMU

- **FAST-LIO `lidar_type 2` per-point time.** Times are taken relative to the header, which is assumed to be the scan start. They are trusted only if the last point's time > 0 (preprocess.cpp).
  - Needs `ts_first_point: true`.
  - Header must equal the first point.
  - Times must run 0–0.1 s.
  - (Verified for our bridge in the 2026-10-21 analysis.)
- **GLIM #257.** Airy divergence was fixed by full frames, `ts_first_point=true`, a single clock, and no double re-timestamping.
- **IMU frame and units.** The Airy IMU is in g, mounted upside down, rotated 90° in z, and has the same frame_id as the cloud (rslidar_sdk #203).
  - Community configs use `R=[0,-1,0,-1,0,0,0,0,-1]`, T=0.
  - **Better: the per-unit DIFOP `imu_calib`** (quaternion + translation). For example, one unit: `T_lidar_imu=[0.00425,0.00418,-0.00446, q(-0.7043,0.7099,0.0013,0.0021)]`. This differs from the generic matrix by about 0.3–0.5° **[U: read our unit]**.
- **Near-range ghosts** (inflated objects, rings) are reported in #181 and are not fixed by return mode or trail filter.
  - Mixed-pixel "trailing" at edges is expected.
  - Use **Strongest** return for mapping.
- **Driver settings.** `dense_points: false` keeps NaNs (our bridge drops them). Use `min_distance` 0.3–0.5 m and `max_distance` about 30 m indoors.
- **Alternative Airy LIO forks:**
  - KiiiLin/fast_lio_robosenseAiry, SaranSundar08/FASTLIO2_robosense_Airy (odometry configs: 0.5/0.5 filters, no dense-map tuning).
  - JunJie1213855/Fast-LIO2_ROS2 (`lidar_type 5`).
  - SuperLIO (blind 1.0).
  - LIO_SC_ROS2 (Point-LIO + ScanContext).

FAST-LIO2 for a dense map

The FAST-LIVO README uses the same parameter semantics.

param	ours	dense indoor
point_filter_num	6	2–3
filter_size_surf	0.5	0.1–0.15
filter_size_map	0.5	0.2–0.3
max_iteration	3	4–5
blind	0.3	0.4–0.6 (operator)
det_range	100	30–60

Offline pose refinement on top of FAST-LIO poses:

- **HBA / BALM2** (HBA): joint plane bundle adjustment. Use indoor voxel 0.5–1 m; the upstream defaults worsened MME in aim-bagr/HBA#4.
- **GLIM** (indoor `voxel_resolution` 0.1–0.25; Airy recipe in #257).
- **Voxel-SLAM** global BA.

Point-cloud post-processing, in order

1. Per-scan range gate [0.4–0.6, 15–20 m].
2. Operator self-filter: a fixed azimuth sector or box in the sensor frame.
3. Jump/mixed-pixel removal on the ring × azimuth image (drop points between neighbours that differ by more than max(5 cm, 3%·r)), plus an incidence angle limit of 75–80°.
4. DUFOMap ray-casting dynamic removal (repo).
5. Glass and reflections: crop window planes, or dual-return reflection detection (STAR-Center).
6. SOR (k 20–30, ratio 1.5–2) followed by a radius filter.
7. Display voxel 3–5 cm, or no voxelising at all: use TSDF meshing.

A clear visual twin

- **Geometry:** TSDF / VDBFusion (voxel 2–3 cm, truncation 3–4×) averages the 1–1.5 cm sensor noise into one surface. Poisson at depth 10–11 (trim low density). NKSR **[U Orin]**.
- **Colour:**
  - Calibrate the extrinsic (direct_visual_lidar_calibration).
  - Estimate the time offset (photometric sweep).
  - Lock exposure and white balance, with a short exposure (the ZED 2i has a rolling shutter).
  - Pick the best view (cos angle × 1/distance × sharpness) instead of averaging.
  - Drop frames with high angular rate.

    - Use a mesh z-buffer for occlusion.
    - Colour mesh vertices or textures.
- **Splats:** the GS-SDF, Gaussian-LIC2 and GS-LIVM family; our `splat/` path (refined poses + LiDAR init + depth loss).
- **Commercial reference:** XGRIDS Lixel uses 2× 48 MP panoramic cameras plus a separate splat pipeline. Visual quality comes from imagery and splatting, not from point colours.
- **FAST-LIVO2** needs a triggered, global-shutter camera; the ZED 2i is neither.

Capture technique

- **Tilt the hemispherical head 45–90°** for floor, ceiling and wall coverage, with the operator in the back half (corridor study PMC9865594; ZEB-style rigs) **[U: angle for Airy]**.
- Start still for 3–5 s.
- Walk at 0.3–0.5 m/s, with rotations under 30–45°/s.
- Cover every room and close loops.
- Hold the rig at a constant offset from your body.
- Pass glass at an angle.
- For the camera, pause 1–2 s per direction.

Metrics without ground truth

- **MME** (mean map entropy, r ≈ 0.3 m indoors), **MPV** (mean plane variance), and plane-thickness RMS on fitted walls. The target is about 1.5–2 cm; worse than 3 cm means a pose, time or extrinsic error.
- Loop-closure gap.
- A/B test each change on the same bag.